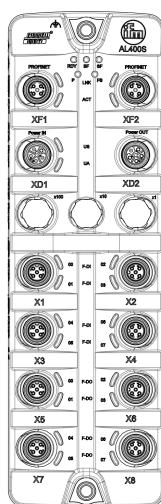


30_200_Testplan_Type_and_System



Testplan Type and System



Project: 153829 Remote_IO_Safe_DIDO_PROFIsafe

Affected Products: eco310 / **AL400S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP67
eco310 / **AL401S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP69K

Status:	 In Review	Confidentiality:	 Internal
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Approved Versions

Current Document version **V1.01** has **not** been approved.

Approved Versions:

- **V1.00** (2025-12-02 14:28)

Review Information Protocol

Purpose: Confirmation of review responsibilities and focus areas by role.

Name	Role	Review Focus
Stephan Mollnau	Functional Safety Manager	Completeness and correct coverage of safety related requirements.
Bastian Ruess	System Architect	Completeness and correct coverage of system-level aspects.
Sebastian Engel	Vice President Technical Management	Overall product functionality of the document are correct.
Ralph Schwarz	Director Product Quality Assurance	Overall product functionality of the document are correct.
Markus Budweiser	Quality	Please check on the quality side whether the formal aspects of the document are correct.

Note: By signing this document, participants confirm that they have reviewed the test plan according to their role and area of focus.



Change Log

(Typ: c = created, m = modified, + = added, - = deleted)

V1.00

Type	Chapter	Author	Description
c	all	D. Topalovic	created a new document
review		D. Topalovic	Invited (B. Ruess, S. Mollnau, J. Burkhardt, M. Budweiser, S. Engel, R. Schwarz)

V1.01

Type	Chapter	Author	Description
m	2.1 7.3	D. Topalovic	Updated to current versions of specification documents updated test specifications
+	8.1	D. Topalovic	Updated DUT Table, added AL400S to DuT Column and updated the calculated amount
+	15.1.5	D. Topalovic	added HW-Version "/ 1.4.0.2" + added link to respective requirement version
+	2.3.3	D. Topalovic	IEC 61131-2:2017 was added
+	15.1.5	D. Topalovic	Verification of power input ports according to IEC 61131-2:2017 (chapter 6.2 ff.) was added
+	2.1	D. Topalovic	Impact of HW Changes from C7 to C7A Document added
+	8.1	D. Topalovic	Added justification for using HW revision 1.4.0.2 for functional testing based on the "Impact of HW Changes from C7 to C7A" document.

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1 General annotations in respect to this document

The first part of this document contains general information about this document, followed by a description of the general part of testing and finally the details of the respective test rounds are discussed.

If several test rounds are required for validation or verification (e.g. additional delta tests) within this project, these are added at the end of the document. For each additional round of testing, a new version of this document should be generated.



2 Scope

This document describes the planning activities for testing (product- and system-level) for the project:

- **153829 Remote_IO_Safe_DIDO_PROFIsafe.**

It defines the objectives and scope of the testing activities.

The following generally applies:

- General information regarding the test strategy [1]

2.1 Applicable documents

Generally, applicable documents can be found here:

Applicable documents

Ref	Document name	Version
[1]	 General information regarding the test strategy	 Eco-Test-24610 - Change history V1.1
[2]	Product requirements Safe_RemoteIO/10_Requirements  - 100_SRIO_Product-Requirements with referred documents in chapter 1.5 in the defined version.	 SRIO-22085 - V1.20
[3]	System requirements Safe_RemoteIO/20_System  - 200_System defines in chapter 1.4 the valid version configuration of the system documents.	 SRIO-15146 - V1.00
[4]	Verification & Validation plan 303_SRIO_Verification_Validation_Plan 	 SRIO-18291 - V1.04
[5]	FSM - Plan 300_SRIO_FSM-Plan 	 SRIO-18290 - V1.05
[6]	Guideline for PROFINET and PROFIBUS PROFIsafe Environment	December 2015 2.6
[7]	House standard Test_specification_Environmental 	 ECO-Standard-2911 - V1.0
[8]	House standard 010_Producttest_EMC 	 ECO-Standard-2909 - V1.1: Note: Editorial changes do NOT lead to a new version.
[9]	Hardware FIT Testspecification 40_301_Hardware_FITs 	 SRIO-20633 - V1.03
[10]	Software FIT Testspecification 40_505_Fault_Injection_Tests 	 SRIO-20426 - V1.02
[11]	Safety Manual https://svn.ifm.com/svn/Safe_Remote_IO/05_UserDocumentation/Safety_Manual	no version exist for now
[12]	Assembly instructions https://svn.ifm.com/svn/Safe_Remote_IO/05_UserDocumentation/Assembly_Instructions	no version exist for now
[13]	Programming manual https://svn.ifm.com/svn/Safe_Remote_IO/05_UserDocumentation/Programming_Manual	no version exist for now



Ref	Document name	Version
[14]	Data sheet https://svn.ifm.com/svn/Safe_Remote_IO/05_UserDocumentation/Datasheet	no version exist for now
[15]	TIA projects for test cases https://svn.ifm.com/svn/Safe_Remote_IO/75_Testautomation/TIA_projects	no version exist for now
[16]	Impact of HW Changes from C7 to C7A 9010_Impact_HW_Changes_C7_C7A 	no version exist for now

Table 2: Applicable documents

2.2 Project specific abbreviations / terms

General abbreviations / terms can be found here:

General abbreviations / terms

Abbreviation	Description
DuT	Device under Test
EMC	Electromagnetic Compatibility
ESD	Electrostatic Discharge
FSM	Functional Safety Management
FAR	Fully anechoic room
FIT	Failure Injection Test
HW	Hardware
IOP	Interoperability
ME	Mechanics
N/A	Not applicable
OS	Operating system
PCB	Printed circuit board
PST	Process safety time
RT	Run time
SW	Software
TIA	Totally Integrated Automation

Table 3: Abbreviations

2.3 Scope of the test planning

The following generally applies:

- General information regarding the test strategy [1]

This test plan covers all tests during the development on product/system level. The tests will be performed on the printed circuit board (PCB) before and after overmolding.



2.3.1 Test mission

[x] New device: A Safe Remote-IO Device which provides 6x2 F-DI and 2x2 F-DO Modules with gateway functionality to PROFIsafe over Profinet

The required SIL 3 and PL e categories must be taken into account during the type approval plan. The safety objective is described in chapter 2.3.3.

- ☐ Corrective action / component replacement
- ☐ Customer specific change
- ☐ Small Software release (with few new features)
- ☐ Small Software release containing bugfixes
- ☐ Software release with extensive modifications
- ☐ Changes in operation system components (Start upper / Bootloader / ifmOS)
- [x] Safety device
- [x] Development from scratch / new device
- ☐ External development
- ☐ Further development
- [x] Safety related functions

2.3.2 Quality targets

When creating the test plan and the consecutive test specifications the following quality targets are to be taken into account:

- ☐ Low return rate
- ☐ Performance
- [x] Robustness (environmental)
- ☐ Precision
- ☐ Power
- [x] Function
- ☐ Power consumption
- ☐ Compatibility
- ☐ Ergonomics
- [x] Documentation
- [x] IOP
- ☐ Changes / new features free of side effects

The quality goals for testing are:

- At least one test case is required for each requirement marked with **"to be tested by: ifm functional test OR Environmental and EMC"**.
- Before execution, approval is required for the test case.
- Verify the correct implementation of each device feature, that the device works as expected and system



requirements are met.

2.3.3 Safety targets

Special focus in test planning shall be put on the verification of the defined measures in the safety concepts and their implementation:

It must be checked at product level whether a device has been developed with the AL4nnS that meets the requirements

- SIL3 according to EN IEC 62061:2021 and EN 61508:2010
- Performance Level e (category 4) according to EN ISO 13849-1:2023
- Validation according to EN ISO 13849-2:2012
- and the EMC requirements according to IEC 61131-2:2017, EN IEC 62061:2021, EN IEC 61000-6-2:2019, EN IEC 61000-6-4:2018 and EN 61326-3-1:2017 (criterion DS).

Safety parameters, such as fault detection should be detected within a specified time (safety concept) and defined reaction.

Special attention in the test planning shall be paid to the verification of the defined measures in the safety concepts and their implementation:

- Verification: 100% coverage of the safety-relevant (SIL Flag) requirements.
- Validation: Correct functioning of the safety functions when requested and under typical fault situations that can be detected by the unit.
- Exclusion of cross-influences of non-safe comfort functions on the safety-relevant product functions.
- Suitability of the user documentation to store, install, commission, configure, program, operate and maintain the safety device within the specified limits.

2.3.4 Development phase according to V-Model

Test level	Product	HW	SW	ME
Type testing (versus requirements)	x			
System test/-integration (versus system test specification)	x			
Module test (versus design/architecture)				
Implementation/Developer test				
Please mark: X = Part of this test plan				

Table 4: Development phase



2.4 Test object

This is the description of the products to be tested or the components to be tested, etc., as well as the assumptions and restrictions that are relevant with regard to the characteristics, the content, and for the scope of the test planning.

2.4.1 Articles handled within this project

The following devices/components are part of the test planning of this test plan:

This test plan covers the products of the Safe Remote-IO Devices family. The family consists of 6x2 F-DI 2x2 F-DO Module.

- AL400S ETH Module PROFI-safe 6x2 F-DI 2x2 F-DO IP67, L-coded power connector)
- AL401S ETH Module PROFI-safe 6x2 F-DI 2x2 F-DO IP69K, L-coded power connector)

The DuT consists of the following components (**the respective version is listed in the test rounds**):

No.	Component	HW	SW	ME	Documentation
1	HW Version (1.4.0.2)	x			
2a	Update container (1.0.0.x)		x		
2b	Bootloader µC3 (4.1.1.257)		x		
2c	Bootloader SCPU's (1.1.0.33)		x		
2d	Option Bytes SCPU's (1.0.0.0)		x		
2e	EEPROM images		x		
3a	Housing food			x	
3b	Housing tool (industry)			x	

Table 5: Components of the DuT(s)

3 Creation, rework, and review of test cases or test programs

The following rules apply to the creation, rework, and review of test cases or test programs, see:

- General information regarding the test strategy - Test cases, test applications and test coverage [1]

3.1 Creation of test cases - general:

General information can be found here:

- Creation of test cases - general [1]

Additionally, when creating a new test case, it is based on the template test case. The template test case brings a structured description and each point of this template description must be checked, but should be deleted if not needed. This improves readability.

All TIA projects used for testing are stored in the SVN under the following link: https://svn.ifm.com/svn/Safe_Remote_IO/75_Testautomation/TIA_projects

If a test case deviates from the standard test case, the specific differences are documented under "Additional Information



for implementation/ Path to the IEC program"

Test cases, which shall run on the test towers, shall be marked with "Automation platform": SET test tower.

Test cases which use device granular passivation shall use the tag "#DeviceGranularPassivation" in the test case.

The default configuration is the "Channel Granular Passivation" - which is not tagged in the test case but mentioned in the special preconditions.

The test case is only tagged if the special case is needed - the Device Granular Passivation.

3.2 Rework of existing test cases

General information can be found here:

- [Rework of existing test cases](#)[1]
- If an already approved test case [which was part of a test run (a pass/fail verdict exists)] needs to be reworked, an issue must be created and the change must be discussed in the CCB meeting. After the CCB agrees on the change, it can be implemented and must undergo a re-review and approval process.

3.3 Review of test cases

General information can be found here:

- [Review of test cases](#) [1]

Before a test case can be set to "Ready for Review", the following custom fields must be defined and completed as part of the review process:

- Automation
- Responsible (department)
- Regression Test
- Severity

The tester who creates the test case is responsible for filling in these fields correctly. The reviewer is responsible for checking that the fields are complete and consistent before approving the test case.

3.3.1 Definition of Custom Fields

There are additional custom fields available, which are optional. However, only those relevant for the test case review are described here.

Automation

- Undecided: Default value – indicates that a decision regarding automation is still pending. Must be changed before review.
- Can be automated: This test can be automated but there is no automation, yet.
- Automated: The test case will be fully executed in an automated way.
- Partly automated: The test case is partly automated and partly executed manually.
- Manual test: The test case will be executed only manually.

If automation is set to "Automated" the custom field "Automation Platform" must be set to the corresponding platform used. Furthermore, all used automation scripts must be versioned in SVN or GitLab and linked in the test case.

Inside SVN, place the script in a subfolder of: \Safe_Remote_IO\75_Testautomation

Inside GitLab, use the following project: <https://gitlab-ee.dev.ifm.ecomatic/sil-3/pqs/functional-test>

In case of automated test cases, the linked scripts are part of the test case review.



Automation: ? Undecided

Execution Platform:

Component: --

Sub-Component: ? Undecided

Description

- ✔ Can be automated
- ✔ Automated
- ✔ Partly automated
- ✔ Manual test

Responsible

- In this test plan, only test cases assigned to ifm functional test or Environmental and EMC test are considered.

Responsible: ifm functional test

Affected Group:

Regression Test: ifm functional test

ifm functional test

Environmental and EMC

There are also other options such as ifm Hardware Test and Software Test. Test cases assigned to these fields are executed by the respective HW or SW teams.

Multiple entries can be selected in the Responsible field if a test case involves more than one responsibility.

Regression Test

- Marked by a check box.
- If checked, the test case belongs to the regression test suite and will be re-executed in regression runs.
- If not checked, the test case will not be part of regression testing.

Regression Test: ☐

Regression Test: ☒

Severity Levels

To organize the test cases, they are divided into three severity levels [default value is "must"]:

When selecting the severity level of a test case, it must correspond to the severity level of the related requirement. If the requirement is classified as a "Must have" the test case is a "must". If the requirement is a "Should have" the test case is a "should". If the requirement is a "Will have" the test case is an "optional".

*Severity: ! must

- ! must
- * should
- + optional

- Must:** These test cases must be executed and passed. They are critical for safety or basic functionality. Skipping them is not allowed.



- **Should:** These test cases should be executed and passed. They are important but not critical. If not performed it has to be reasoned.
- **Optional:** These test cases are not required to the verification of safety or critical functions, but improve the overall test coverage.

3.4 Test coverage

General information on test coverage can be found here:

- [General information regarding the test strategy - Review process and test coverage](#)
- The review participant suggestion can be found there, too
- The test case coverage of all requirements is ensured through the defined review process.
The corresponding role distribution is listed in a table at the beginning of each document.
Each participating role verifies, according to its area of responsibility, that all relevant requirements are completely and correctly covered by appropriate test cases.
- All requirements assigned to "to be tested by: ifm functional test", "Environmental and EMC", "ifm hardware test" or "ifm software test" and not marked as Draft or Rejected, are listed in the test coverage overview.
- For each requirement assigned to "to be tested by: ifm functional test", "Environmental and EMC", "ifm hardware test" or "ifm software test" it is verified whether a corresponding test case is linked. This guarantees that every valid requirement is covered by at least one test case.

There are the following resources to verify test coverage:

The Requirement to Test Case per Document overview can be accessed via the following link:  [Test Coverage: Requirement To Test Case Per Document](#). This report allows you to select specific documents and displays all relevant requirements assigned to ifm functional test, Environmental and EMC, ifm hardware test or ifm software test, helping you verify whether each requirement is linked to at least one test case.

The Test Coverage Project Overview is available here:  [Test Coverage Project Overview](#). It provides a complete overview of all relevant documents and their associated test case coverage.

3.5 Test applications

General information can be found here:

- [Test applications](#) [1]
- The TIA test applications can all be found in the SVN repository at: https://svn.ifm.com/svn/Safe_Remote_IO/75_Testautomation/TIA_projects



4 Test methods

The general descriptions of the individual test methods can be found in the following document, see:

- [General information regarding the test strategy - Test methods \[1\]](#)

The respective test methods are described in Chapter 7.3 Test specification, where all test specifications are listed. For each test specification, the corresponding test method is defined. In the document, the respective test method is indicated at the beginning of each test specification.

5 Test cycles

For test cycles the following rules apply, see:

[General information regarding the test strategy - Test cycles \[1\]](#)

See also [15 - Test strategy](#)

6 Limitations and assumptions

6.1 General limitations and assumptions

The following restrictions and assumptions apply to all testing activities, see:

- [General information regarding the test strategy - General limitations and assumptions \[1\]](#)

6.2 Project specific limitations and assumptions

- On fieldbus side PQS selected the following PLC's:
 - The focus is on Siemens PLC S7-1511F (market leader for PROFI-safe)
 - Compatibility tests with the Siemens PLC S7-12nn
 - Compatibility tests with the Phoenix PLC RFC4072S (PLCnext)
- The difference between the two variants lies in the material of the sleeve: The AL400S has Nickel-plated brass sleeves and the AL401S has stainless steel sleeves. The allocation of the two variants, specifying which derivative is used for EMC and environmental testing, is discussed in the related issue.  [SRIO-18289 - Selection of the devices for EMC and environmental - needs confirmation](#)
However, the ingress protection (IP) tests are carried out on both variants to validate the sealing performance.
- The quality of the test cases is only as good as the quality of the requirements.
- The completeness of the test cases is only as good as the completeness of the requirements.
- Regression tests and the long-term test should be defined and as much as possible automated.
- The long-term test should be performed with the agreed duration of at least 7 days with the final FW.
- If there are security expectations, these are specified via requirements.
- If functional, environmental and EMC testing is not carried out on near-series equipment, a change impact analysis must be performed to evaluate the quality of the test results.
- Profinet/PROFI-safe qualification tests will not be done in the context of this test plan and is carried out by a certified laboratory.
- Software FITs and HW FITs on module level (close to unit test) will be performed, but are not part of this test plan



[9] [10].

- UL approval is performed externally after type testing.
- EMC to show compliance to FCC/ISED.
- DNV is not included in this project.
- Salt mist test covered by NEMA 250 5.13.1
- Mounting on DIN rail is a use case to be considered for device temperature tests.
- Tests during development don't belong to the type and system test.
- An essential device configuration is the selection of the profisafe module - which can be either device granular passivation or channel granular passivation. All tests will be done with channel granular passivation. Only selected tests will be retested with device granular passivation. These tests will be marked in the test case with "#DeviceGranularPassivation" and the TIA project for the test must be linked and saved in SVN.

Description: Difference between those modules is the F_Passivation parameter, which affects the error handling of the device, if one or all channels are deactivated. For device granular passivation (legacy setting) in case of an error all channels are deactivated, for the channel granular passivation it will depend on the error but generally the goal is to deactivate only the faulty channel. Since in both cases the error reaction of the module is performed and the only difference is the availability of the other channels, the test activities will focus on the channel granular setting and only partly on the device granular setting if explicitly needed for the test.

7 Test documentation

General information on test documentation (safety plan/test plan/test specification/etc.) can be found here:

- [General information regarding the test strategy - Test documentation](#) [1]

7.1 Safety-plan (if applicable)

For this project the following safety plan applies:

- [300_SRIO_FSM-Plan](#) [5]

7.2 Test plan

General information can be found here:

- [Test plan](#) [1]

7.3 Test specifications

General information can be found here:

- [Test specifications](#) [1]

The applicable test specifications can be found below. Each test specification includes the corresponding test method.

Test specification	Test methods
30_100_EMC_test_specification	Functional, blackbox, stress and ambient tests
30_112_Environmental_test_specification	Environmental tests



Test specification	Test methods
 30_110_GeneralProductInformation_test_specification	Formal and document review
 30_101_IOP_test_specification	Functional and blackbox tests
 30_210_Power_test_specification	Functional, blackbox and stress tests
 30_211_Digital_IO_test_specification	Functional and blackbox tests
 30_212_Communication_Profinet_test_specification	Functional, blackbox and stress tests
 30_213_IoTCore_test_specification	Functional, blackbox and stress tests
 30_214_SCPUs_test_specification	Functional, blackbox tests
 30_215_FW-Update_test_specification	Functional and blackbox tests
 30_216_ZVEI_test_specification	Functional, blackbox and performance tests
 30_217_SET_Test_Tower	Functional, blackbox, performance, stress, FIT tests
 30_218_Operating Modes_test_specification	Functional, blackbox tests
 30_219_Timing Constraints_test_specification	Functional, blackbox and performance tests
 30_2110_Long_Term_Test	Functional, blackbox and performance tests
 30_220_Communication_Profisaftest_Specification	Functional, blackbox tests
 30_222_Diagnostic_test_specification	Functional, blackbox tests

7.4 Test case descriptions

General information can be found here:

- [Test case description](#)[1]

7.5 Test protocol

General information can be found here:

- [Test protocol](#) [1]

see the link for the naming convention of test runs

7.6 Test report

The corresponding test report will be found in Polarion under the following link:

- [Test report](#) [1]
-  [153829_Test_report_master](#)



8 Test resources

8.1 Devices under test, amount, configurations

At least the following test devices are required for the testing activities (see chapter "Test object"):

According to the Impact of HW changes from C7 to C7A [16], there are no functional changes between HW revision 1.4.0.2 (C7) and the current hardware revision (C7A) affecting the scope of the functional tests. Therefore, the functional tests will be performed using HW revision 1.4.0.2 (C7).

DuT	HW revision / version	Test area	Amount
AL400S and AL401S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2 Food: LPB: 80343809_03 LPB SAFE_RIO_6DI_2DO_FOOD HW1.4.0.2	Environmental tests internal	TS1 (temperature cycle test and IP X7 and X9K): 6 (3x AL400S and 3x AL401S) (remark: 6X test is done externally) TS3 (damp / heat test): 3 (only external) TS4 (component and device temperature tests): 1x AL400S + 1 board AL400S TS5 (vibration and IP X7 and X9K): 6 (3x AL400S and 3x AL401S) TS6 (free fall): 3 (3x AL401S)
AL400S and AL401S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2 Food: LPB: 80343809_03 LPB SAFE_RIO_6DI_2DO_FOOD HW1.4.0.2	Environmental tests external	Environmental external: 18 (9x AL400S and 9x AL401S)
AL400S and AL401S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2 Food: LPB: 80343809_03 LPB SAFE_RIO_6DI_2DO_FOOD HW1.4.0.2	Impact test internal according to EN (IEC) 61010-1/-2-201	6 (3x AL400S and 3x AL401S)
AL400S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2	EMC with FCC/ISED	EMC external: 10 (10x AL400S) (precondition: if no DUTs break)
AL400S and AL401S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2 Food: LPB: 80343809_03 LPB SAFE_RIO_6DI_2DO_FOOD HW1.4.0.2	Functional test safety / functional test non-safety / stress test / performance test / HW/ electrical test / FW-Update test	20 boards (10x PCB AL400S and 10x PCB AL401S) (to have some additional devices)



DuT	HW revision / version	Test area	Amount
AL400S and AL401S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2 Food: LPB: 80343809_03 LPB SAFE_RIO_6DI_2DO_FOOD HW1.4.0.2	Long-term tests	5 (5x AL400S)
AL400S	The following hardware version is planned: Industrie: LPB: 80343808_03 LPB SAFE_RIO_6DI_2DO HW1.4.0.2	Automation	4x AL400S or AL401S without connector

Table 6: Required amount of DUT's

The details of the test devices (e.g. serial number or similar identification feature) are documented in the test run when the test is carried out. All relevant versions, including the hardware, software and component versions (as listed in Chapter 2.4.1) are specified in the respective test run.

8.2 Testers and roles

Testing is planned to be done by the following persons:

Role / Task	Assignees
Test engineer environmental tests	Norbert Schwarz, Rüdiger Braetsch
Test engineer EMC tests (CE)	Norbert Schwarz
Test engineer functional	Klaus Kaiser, Till Monzel, Dragana Topalovic
Test manager	Dragana Topalovic

Table 7: Testers

9 Effort

9.1 General effort

General information on the preparation of test activities, such as introduction, the creation of test set-ups or effort for test implementation, can be found here:

- [General effort](#)

9.2 Project specific effort

- Free testing of new firmware releases since March 2025
- All test specifications completed and reviewed by the end of November 2025
- Setup to be prepared for external EMC tests by begin of January 2026
- Setup for prepared for external Environmental tests by begin of January 2026
- Setup for Long term test by begin of December 2025
- Development of testautomation by end of December 2025
- Retesting of open issues on going
- Test run management and execution for every release candidate
- Start first test run by mid of November 2025



- Bugfixing on going

Software releases

The software is developed in modules and delivered to us in regular releases. About every three weeks we receive a new release where certain modules are finished and ready for testing. These modules are then used for targeted verification of the available features and development of the related tests.

Goal: Before starting the full-test, we expect at least 50% test coverage on the released modules.

Test coverage is measured in the amount of requirements of the system architecture, that are covered with test cases that are in status approved, vs the total amount of requirements in the system architecture.

The releases are listed in the following link: [ReleasePlanung](#)

10 Schedule

The following generally applies, see:

- [General information regarding the test strategy - Schedule \[1\]](#)
- [SRIOProjectPlans](#)

11 Test environment

The following generally applies to test environments, see:

- [General information regarding the test strategy - Test environment \[1\]](#)

All environmental tests specified in the datasheet will be carried out externally.

11.1 Automatic execution of test cases - test equipment

11.1.1 Nightly Build test tower (only development)

Before the software is handed over to the test team, a set of automated basic tests is run on the nightly build test tower. These include checks like communication startup and the basic behavior of inputs and outputs with different parameter settings.

These tests help find major issues early, so they can be fixed before formal testing begins.

This is a development-level test only and is not part of the type or system testing.

A detailed description of the nightly build test tower and the software involved can be found here:

[Nightly Build Test Tower < Testing < Documents & Pages < Safe_RemotelO](#)

11.1.2 Long-term Test

The long-term test is a small test setup with a PLC, and a small set of devices. The goal is to run the devices over a extended amount of time and check if they are stable or run into errors over time.

A detailed description of the continuous test and the software involved can be found here:

[Long Term Test < Testing < Documents & Pages < Safe_RemotelO](#)



11.1.3 SET Test Tower I&II

Type	Description	Inventory number	Calibrated until
SET Test tower I (sRIO001)	The SET test tower includes two power supplies, four electronic loads, one industrial switch, one Siemens PLC, FPGA, digital multimeter, oscilloscope.	12612	April/May 2026 Safe_Remote_IO - Revision 3555: /75_Testautomation/Kalibrierung
SET test tower II (sRIO002)	The SET test tower includes two power supplies, four electronic loads, one industrial switch, one Siemens PLC, FPGA, digital multimeter, oscilloscope.	12613	April/May 2026 Safe_Remote_IO - Revision 3555: /75_Testautomation/Kalibrierung

Table 8: Details of test equipment

Automated tests for functional type- and system-tests will be run on the so called SET test towers “sRIO001” and “sRIO002”. The towers contact one DUT each, via contact pins in a needle bed adapter.

Tests to be run on the SET towers are, for example, device parameter configuration tests (all iParameters and F-Parameters) as well as FITs.

Depending on the time available there are many options to further add tests to the automation and to increase the amount of regression tests. The two SET towers are the main test automation setup.

A detailed description of the tests done on the SET towers as well as the software and hardware involved can be found here:

[SET Test Tower < Testing < Documents & Pages < Safe_RemoteIO](#)

11.2 Manual execution of test cases

The following generally applies to manual tests, see:

- [General information regarding the test strategy - Manual execution of test cases \[1\]](#)

11.3 Tools and utilities in safety projects

The following generally applies to tools and utilities in case of safety projects, see:

[General information regarding the test strategy - Tools in safety projects \[1\]](#)

The following tools / utilities are required for test execution (for version see [301_SRIO_FSM_Tool-Qualification](#))

Type	Description
LabVIEW	Graphical programming language and environment. Only used for the SET test towers i.e. automated tests.
NI Teststand	Test framework. Only used for the SET test towers i.e. automated tests.
TIA Portal	Siemens PLC programming. For manual and automated tests. All device configurations are done with this tool.



Type	Description
PLCnext	Phoenix PLC programming. Only manual test to verify compatibility with this tool.
Visualizer	IoT Core access.
ifm-CRC-Tool	Tool to calculate iPar-CRC for device configuration.

Table 9: Tools and utilities

12 Assessment of the test activities

General information on test performance assessment can be found here:

- [General information regarding the test strategy - Assessment of the test activities](#) [1]

12.1 Evaluation of test results

12.1.1 Pass/fail criteria

In general, the following risks apply, which can affect or influence the test planning:

- [Evaluation of test results - pass / fail criteria](#) [1]

12.2 Handling of issues/change requests

The issues/change requests in this project will be handled by:

- Issues/change requests created in Polarion.
- These issues/change requests will be discussed and resolved during CCB meetings.
- In case of none-safety relevant Issues/change requests the CCB is able to decide to leave topics open by setting the resolution "later" which means those topics will be shifted to the next project / outside of the scope of this project. Safety relevant issues and changerequests has to be solved. See: [Issues and change requests - later](#)

see all open issues and change requests: [Issues and change requests - open](#)

see all resolved issues and change requests: [Issues and change requests - resolved](#)

13 Risks of test planning / execution

In general, the following risks apply, which can affect or influence the test planning:

- [General information regarding the test strategy - Risks of the test planning / execution](#) [1]

To access the risk register, open the provided link and select "Risk Register" from the top navigaton bar

- [Project: Information V2: Remote_IO_Safe_DIDO_PROFIsafe - Power Apps](#)



14 Remarks

14.1 Traceability

General information on traceability can be found here:

General information regarding the test strategy - Traceability [1]

15 Test strategy

The assignment of the individual validation steps to the test methods and their explanations can be found in the following chapters.

- 15.1.2 - Applied test methods in the first test run
- 15.1.5 - Overview of planned test areas

15.1 First Full Test Round

15.1.1 Validation / verification

N/A

15.1.1.1 Unchanged functionality

N/A

15.1.1.2 New / changed functionality

N/A

15.1.1.3 Bug fixes

N/A

15.1.1.4 Regression tests

N/A

15.1.1.5 Use cases

Planned to be executed are the following tests:

- Use case tests (long-term test)
- Combined long-term and stress
- Selected performance tests



15.1.1.6 Environmental / mechanical tests

New device - environmental test carried out partly, where needed.

15.1.2 Applied test methods in the first test run

Generally applicable and further information related to test methods can be found here:

- [General information regarding the test strategy - Test methods \[1\]](#)

The following marked methods are intended for validation at product level:

Add reference to safety plan in case of safety project

No.	Method:	System level:
1	Functional or Blackbox tests	x
2	Performance or stress tests	x
3	Formal review	
4	Statistical test methods	
5	Ambient, environmental tests	x
6	Fault insertion tests	
7	Document review	x
8	Model based testing	

Table 10: Test methods

15.1.3 Variant selection

The PROFIsafe variants AL400S and AL401S are selected for this test plan.

15.1.3.1 Hardware differences

AL400S: Nickel-plated brass sleeves

AL401S: stainless steel sleeves

The EEPROM content (article_name and order_number) differs between AL400S and AL401S.

15.1.3.2 Software differences

None

15.1.3.3 Conclusion

N/A

Remark: If necessary, this conclusion can also be applied to issues and bug fixes.



15.1.4 Measures to achieve the test targets

The test strategy describes which techniques are used for validation / verification. The choice of techniques takes into account the test object, the assumptions and limitations regarding the test object, and the relevant phase in the V-Modell.

The following tests can be derived from the test strategy considerations:

15.1.4.1 Validation / verification of unchanged functionality

See [15.1.1.1 - Unchanged functionality](#)

15.1.4.2 Validation / verification of new / changed functionality

See [15.1.1.2 - New / changed functionality](#)

15.1.4.3 Validation of bug fixes

See [15.1.1.3 - Bug fixes](#)

15.1.4.4 Regression tests

See [15.1.1.4 - Regression tests](#)

15.1.4.5 Validation of use cases

The following test case catalogues are to be used for the applicability in use case applications

N/A

15.1.4.6 EMC tests

New device - EMC test must be carried out to the full extent.

15.1.4.7 Environmental (incl. mechanical) tests

N/A

15.1.5 Overview of planned test areas

The test strategy describes which techniques are used for validation. The choice of techniques takes into account the test object, the assumptions and constraints regarding the test object, and the relevant phase in the V-model.

The following test areas and tests can be derived from the test strategy considerations:

No	Technology / Measure	Target to achieve	DUT variant / HW version	Requirement specification version	Test location
1a	(Environmental test) Dry heat test according to IEC 61131-2:2017 (chapter 5.2.1 table 4)	Verify robustness against temperature.	AL400S and AL401S / 1.4.0.2	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
1b	(Environmental test) Cold test according to IEC 61131-2:2017 (chapter 5.2.1 table 5)	Verify robustness against temperature.	AL400S and AL401S / 1.4.0.2	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025



No	Technology / Measure	Target to achieve	DUT variant / HW version	Requirement specification version	Test location
2a	(Environmental test) Temperature cycle test according to DIN EN 60068-2-14 and IEC 61131-2:2017 (chapter 5.2.1 table 6) low temperature limit	Verify robustness against temperature cycles.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
2b	Temperature shock according to IEC 61131-2:2017 (chapter 5.2.1 table 6)	Verify robustness against fast temperature cycles.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
3a	(Environmental test) Damp heat constant according to EN 60068-2-78	Verify the robustness against humidity.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
3b	(Environmental test) Damp heat cyclic according to IEC 61131-2:2017 (chapter 5.2.1 table 7)	Verify the robustness against humidity.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
4	(Environmental test) Sinusoidal vibration according to IEC 61131-2:2017 (chapter 5.3.2 table 9 and 10)	Verify robustness against vibration.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
5	(Environmental test) Mechanical shock according to IEC 61131-2:2017 (chapter 5.3.3 table 11)	Verify robustness against mechanical shock.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
6	(Environmental test) Free fall with packaging according to IEC 61131-2:2017 (chapter 5.4.4 table 15)	Verify packaging protection and device robustness against free fall.	AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
7a	IP 6X, IP X5, IP X6 and IP X7 according to DIN EN 60529 + Corr. 1:2017 + Corr. 2:2019 Remark: IP X5 is covered by IP X6 and will not be verified separately.	Verify robustness against ingress of dust and water.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
7b	IP X9K according to DIN EN 60529 + Corr. 1:2017 + Corr. 2:2019	Verify the robustness against the ingress of high-pressure water.	AL401S / 1.4.0.2	SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
8	(Environmental test) Test sequence 1 Temperature cycle test and IP X5, IP X7 and X9K according to the inhouse standards [7]	Verify robustness against temperature cycles.	AL401S / 1.4.0.2	SRIO-15146 - V1.00	Internal
9	(Environmental test) Test sequence 4 Component and device temperature tests according to the inhouse standards [7]	Ensure components operate within the specification limits. Surface temperatures remain within the permitted range.	AL401S / 1.4.0.2	SRIO-15146 - V1.00	Internal and TÜV-Witnessing
10a	(Environmental test) Test sequence 5 Vibration according to inhouse standard	Verify robustness against vibration.	AL400S and AL401S / 1.4.0.2	SRIO-15146 - V1.00	Internal
10b	IP6X*, IP X5, IP X6 and IP X7 according to inhouse standard Remark: IP X5 is covered by IP X6 and will not be verified separately.	Verify robustness against ingress of dust and water.	AL400S (same DUT as used in 10a) / 1.4.0.2	SRIO-15146 - V1.00	Internal IP6X* external, as internal is not possible



No	Technology / Measure	Target to achieve	DUT variant / HW version	Requirement specification version	Test location
10c	IP6X*, IP X5, IP X6, IP X7 and IP X9K according to inhouse standard	Verify the robustness against the ingress of high-pressure dust and water.	AL401S (same DUT as used in 10a) / 1.4.0.2	 SRIO-15146 - V1.00	Internal IP6X* external, as internal is not possible
11	(Environmental test) Test sequence 6 Free fall without packaging according to inhouse standard	Verify packaging protection and device robustness against free fall.	AL401S / 1.4.0.2	 SRIO-15146 - V1.00	Internal
12	EMC tests according to IEC 61131-2:2017, EN IEC 61000-6-2 2019-11 (immunity), EN IEC 61000-6-4 2020-09 (emission) and EN 61326-3-1 2018-04 (functional safety)	Validate of the EMC immunity and emission requirements.	AL400S / 1.4.0.2	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
13	Verification of power input ports according to IEC 61131-2:2017 (chapter 6.2 ff.)	Verify robustness against power input ports.	AL400S / 1.4.0.2	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
14	FCC/ISED conformity	Conformity Class A	AL400S	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025 at EMCE GmbH
15	Chemical resistance according to ISO 16750-5 and NEMA 250 5.13.1	Verify resistance against cleaning agents and chemicals.	AL400S and AL401S	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
16	Ecolab test method 40-1 according to DIN EN ISO 175	Verify resistance against Ecolab cleaning agents according to test method 40-1.	AL400S and AL401S	 SRIO-15146 - V1.00	Externally accredited laboratory according to EN ISO/IEC 17025
17	Impact test according to EN (IEC) 61010-1/-2-201 (chapter 8.2.2)	Verify mechanical robustness of the device housing against impact.	AL400S and AL401S / 1.4.0.2	 SRIO-15146 - V1.00	Internal
18	Functional test safe - Standard functional test - Stress test - Performance test - Profisafe	Verify correct functional behavior under normal and stress conditions, covering requirements regarding safe device functions on product and system level.	AL400S / 1.4.0.2	 SRIO-15146 - V1.00	Internal
19	Functional test non-safe - IoT Core - Update - Profinet	Verify non-safe functional behavior.	AL400S / 1.4.0.2	 SRIO-15146 - V1.00	Internal
20	Long-term test (at least 7 days)	Verify long-term stability of device operation	AL400S and AL401S / 1.4.0.2	 SRIO-15146 - V1.00	Internal



No	Technology / Measure	Target to achieve	DUT variant / HW version	Requirement specification version	Test location
21	FW-Update test	Verify firmware update functionality and robustness	AL400S	SRIO-15146 - V1.00	Internal
22	Functional FIT	Verify detection of HW faults which result in SW reactions. (Is marked to be tested by: ifm functional test)	AL400S / 1.4.0.2	SRIO-15146 - V1.00	Internal
23	User documentation review	Ensure completeness, correctness and usability of the user documentation (accompanying documentation, operating manual, assembly instructions, programming manual, data sheet) based on requirements	AL400S and AL401S	SRIO-15146 - V1.00	Internal
24	Netload-Class-III Test	Independent external validation according to Netload-Class-III requirements	AL400S / 1.4.0.2	SRIO-15146 - V1.00	External at IFAK
25	Profinet Certification	Verify compliance with Profinet specification and certification requirements.	AL400S / 1.4.0.2	SRIO-15146 - V1.00	External at IFAK
26	PROFIsafe Certification	Verify compliance with PROFIsafe specification and certification requirements.	AL400S / 1.4.0.2	SRIO-15146 - V1.00	External at IFAK

Table 11: Overview of planned test areas

15.2 Second test run (full test/ delta test)

Note: If another round of testing is required, the testing strategy shall be documented hereafter similar to chapter 15.1.

Test round	Test specification	Test Scope	FW Version	Tester	Comment
TR-1	30_212_Communication_Profinet	Full	FW 1.1.1.873	Hajer Mouldi	



Test round	Test specification	Test Scope	FW Version	Tester	Comment
TR-1	30_212_Communication_PROFIsafe	Full	FW 1.1.3.877	Rainer Sonnenmoser	
TR-1	30_218_Operating Modes	Full	FW 1.1.1.873	Till Monzel	
TR-1	30_210_Power	Full	FW 0.5.0.830	Klaus Kaiser	
TR-1	30_213_IoTCore	Full	FW 1.1.1.873	Alexander Meier	
TR-1	30_215_FW-Update	Full	planned in KW 31	Alexander Meier	
TR-1	30_222_Diagnostic	Full	FW 1.1.3.877	Marcus Walther	
TR-1	30_216_ZVEI	Full	planned in KW 31	Alexander Meier	

- End of document -